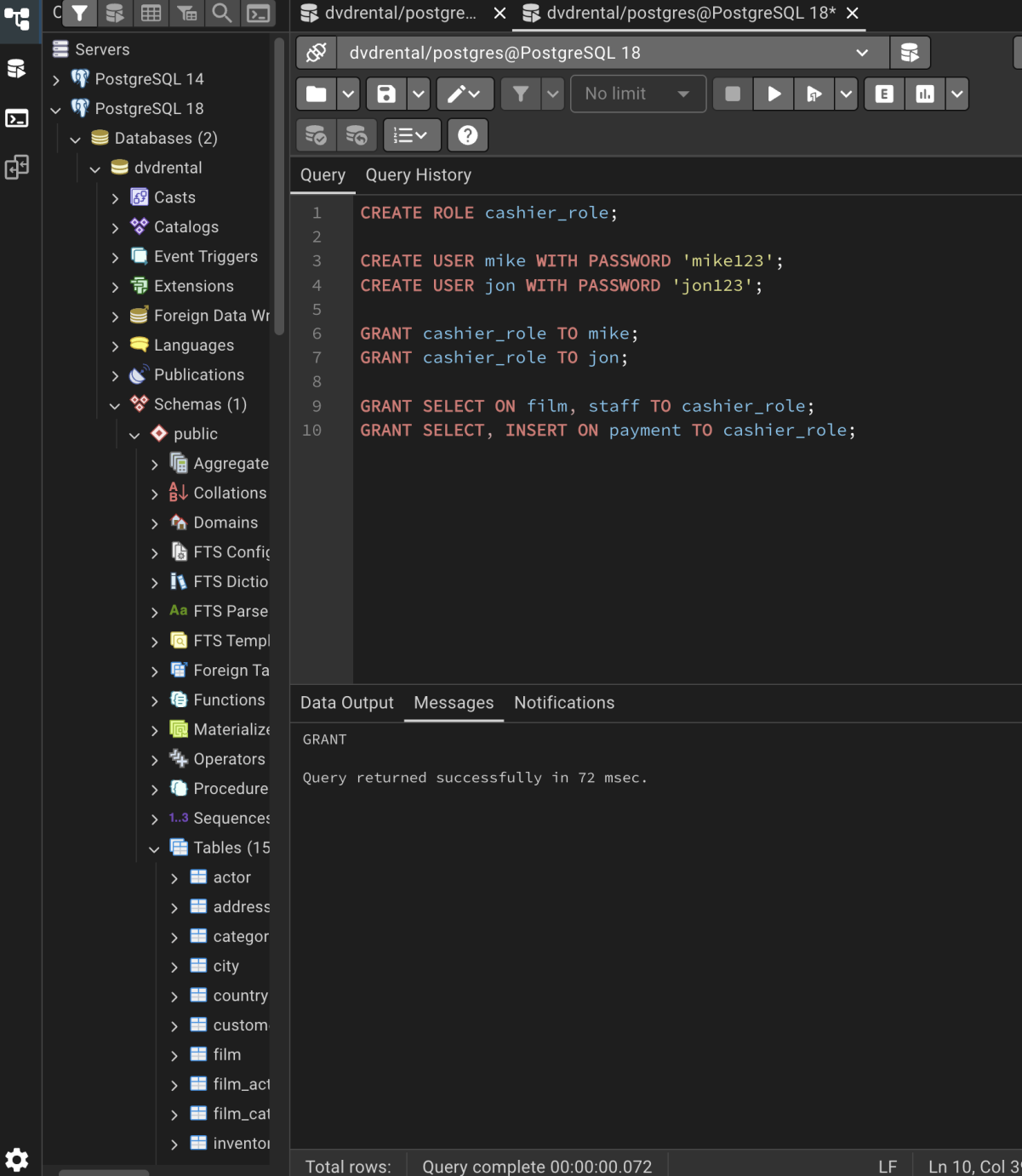


Lab 1: Implementing RBAC (Solution)



The screenshot displays a PostgreSQL client interface with a dark theme. On the left, a sidebar shows the database structure: Servers (PostgreSQL 14, PostgreSQL 18), Databases (2) (dvdrental), and Schemas (1) (public). The 'dvdrental' database is selected, showing its contents: Casts, Catalogs, Event Triggers, Extensions, Foreign Data Wrappers, Languages, Publications, and Schemas (1) (public). The 'public' schema is expanded, showing various database objects like Aggregate Functions, Collations, Domains, FTS Configuration, FTS Dictionaries, FTS Parsers, FTS Templates, Foreign Tables, Functions, Materialized Views, Operators, Procedures, Sequences, and Tables (15). The 'Tables' folder is expanded, showing tables like actor, address, category, city, country, customer, film, film_actor, film_category, and inventory.

The main window shows the 'Query' tab with the following SQL script:

```
1 CREATE ROLE cashier_role;
2
3 CREATE USER mike WITH PASSWORD 'mike123';
4 CREATE USER jon WITH PASSWORD 'jon123';
5
6 GRANT cashier_role TO mike;
7 GRANT cashier_role TO jon;
8
9 GRANT SELECT ON film, staff TO cashier_role;
10 GRANT SELECT, INSERT ON payment TO cashier_role;
```

The 'Query History' tab is also visible, showing the same query. Below the query editor, the 'Data Output' tab is selected, displaying the message: 'GRANT' and 'Query returned successfully in 72 msec.'.

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.072' and 'Ln 10, Col 39'.

Lab 2: Implementing RLS (Task)

The screenshot shows a PostgreSQL IDE interface with the following components:

- Left Panel (Servers):** Displays a tree view of the database structure. The 'dvdrental' database is selected, showing its various components like Casts, Catalogs, Event Triggers, Extensions, Foreign Data Wrappers, Languages, Publications, and Schemas. The 'public' schema is expanded, listing tables such as actor, address, category, city, country, custom, film, film_actor, film_category, and inventory.
- Top Panel (Query Editor):** Shows the connection 'dvdrental/postgres@PostgreSQL 18'. The query editor contains the following SQL code:

```
2  
3 CREATE USER mike WITH PASSWORD 'mike123';  
4 CREATE USER jon WITH PASSWORD 'jon123';  
5  
6 GRANT cashier_role TO mike;  
7 GRANT cashier_role TO jon;  
8  
9 GRANT SELECT ON film, staff TO cashier_role;  
10 GRANT SELECT, INSERT ON payment TO cashier_role;  
11  
12  
13 ALTER TABLE payment ENABLE ROW LEVEL SECURITY;  
14  
15 CREATE POLICY isolate_cashier_payments ON payment  
16 FOR SELECT  
17 TO cashier_role  
18 USING (  
19     staff_id = (SELECT staff_id FROM staff WHERE username = curr  
20 );
```

An 'Execute query' tooltip is visible over the 'Execute' button (a play icon) in the toolbar.

- Bottom Panel (Messages):** Displays the execution results. The 'Messages' tab is active, showing the message: 'Query returned successfully in 30 msec.' The 'Data Output' tab is also visible but empty.
- Status Bar:** At the bottom, it shows 'Total rows: Query complete 00:00:00.030' and 'LF Ln 13, Col 1'.